Htrack26 Project Progress Report

Work completed on 2 November 2026

1. Introduction

The work carried out today focused on connecting the Htrack26 Flutter application to a new Supabase backend and setting up the first part of user authentication. The application was tested on an Android emulator.

2. Development environment and packages

The project was opened from C:\src\Health-Tracking-App. The supabase_flutter package was already included in the Flutter dependencies. The dependency command completed successfully. Notices about newer package versions were informational and did not prevent the application from running.

3. Supabase project connection

A new Supabase project was used for the application. The project URL was identified from the Supabase connection details and Data API page. The REST endpoint included /rest/v1/, but Flutter initialization uses the base project URL without that suffix. The app was initialized with the project URL and a publishable key. The secret key was not required in the Flutter client.

4. Database setup

A connection_test table was created and populated with a sample message. A query from Flutter successfully read the row and printed the result in the VS Code terminal. This confirmed that the application could communicate with the Supabase database.

5. User profiles and database security

A profiles table was created. Its id column is a UUID linked to auth.users, and it stores the user's full name, role, and creation time. The allowed roles are patient, doctor, and researcher. Row Level Security was enabled, and a select policy was added so an authenticated user can read only the profile with the same user ID. A database function and trigger were added to create a profile automatically when a new Auth user is registered. New public registrations are assigned the patient role by default.

6. Authentication screen

A new file named auth_screen.dart was created. It contains a basic registration and login interface. The registration form collects a full name, email address, and password. The login form uses an email address and password. Basic form validation was added, and authentication errors are displayed on the screen. Registration uses Supabase Auth signUp and sends the full

name as user metadata. Login uses signInWithPassword. A sign-out action was also added. After successful login, the current version displays a temporary login-success screen with the signed-in email address.

7. Testing and results

The application launched successfully after Supabase initialization. The database connection test returned the expected sample message. A test user was able to log in. The user was visible in Supabase Authentication → Users, and a corresponding record appeared in the profiles table. The automatic profile creation trigger therefore worked for the tested registration.

8. Email rate-limit issue

During initial registration testing, Supabase returned an email rate-limit message after several attempts. This was related to the email sending limit of the default service. Email confirmation was temporarily disabled for development testing. A production email provider and appropriate confirmation settings will be needed before public release.

9. Current status

The Supabase client setup, database read test, basic authentication operations, profiles table, RLS select policy, and automatic patient profile creation have been implemented and tested. The existing patient dashboard has not yet been connected to the authentication flow. Doctor and researcher access, health-record management, and production security policies remain to be implemented.

10. Planned next steps

The next work will be to preserve the existing dashboard code while keeping authentication in a separate file; add session-aware navigation; load the user's profile and role after login; connect patient accounts to the existing patient dashboard; create an approval process for doctor and researcher accounts; design health-record and doctor-patient relationship tables with appropriate RLS policies; plan anonymized researcher access; test the complete authentication and permission flow; and review the repository for secrets before pushing the changes to GitHub.

11. Conclusion

The main achievement of the day was establishing a working connection between the Flutter application and Supabase. The first authentication flow and automatic patient profile creation were also tested successfully. The next stage is to connect authentication with the existing dashboard and build the role-based features with proper access controls.

Prepared as a record of the development work completed for Htrack26.